

3.6 Doing It Many Times

while - repeat until told to stop

```
while (true) {  
    std::print("Friend's name: ");  
    std::string friendName;  
    std::getline(std::cin, friendName);  
  
    if (friendName.empty()) {  
        break; // empty line means we're done  
    }  
  
    friends.push_back(friendName);  
}
```

A collection, ready-made

C++ ships with ready-made collections - you don't have to build your own growable list.

```
std::vector<std::string> friends;
```

`std::vector<std::string>` is a list of strings that **grows** as you add to it with `push_back`.

Same idea as `std::string` being a ready-made collection of characters.

Looping over the collection

```
std::println("\nYou added {} friend(s)", friends.size());  
for (const std::string& friendName : friends) {  
    std::println(" - {}", friendName);  
}
```

A range-based `for` visits every element, one at a time.

break

`break` exits a loop immediately, regardless of the loop's condition.

```
if (friendName.empty()) {  
    break;  
}
```

Without it, `while (true)` would never stop on its own.

Next: 3.7

Saving and Loading Files